

STADIUMPULSE AI

GenAI Command Center for Global Soccer Tournament 2026

Unified fan experience, operational intelligence, and strategic overview.

Project Overview & Documentation

This document contains the project elevator pitch, interactive demo script, architectural blueprint, 8-pillar alignment matrix, security specifications, and development setup details for StadiumPulse AI.

Document Metadata

Created For:	PromptWars Challenge 4 — Stadium Operations Category
Author Role:	Senior Full-Stack Observability & Security Lead
Technologies:	Next.js 14, Gemini 2.0 Flash, Tailwind CSS, Vitest, Zod
Target Stadium:	MetLife Stadium, NJ (Fictional Tournament Host)
Release Date:	July 11, 2026

1. EXECUTIVE SUMMARY & PITCH

The 30-Second Elevator Pitch:

Managing 80,000 fans at a Global Soccer Tournament stadium is a complex coordination challenge. Fragmented systems leave fans lost, language barriers unresolved, and operational staff reacting too late to critical incidents. StadiumPulse AI is a unified, GenAI-powered command center. We have built two seamless portals: a Fan Portal for intuitive wayfinding, multilingual chat, and transit ETAs, and an Ops Command Center that turns raw crowd data into proactive shift briefings and incident action plans. By replacing static maps and slow manuals with Gemini 2.0's real-time reasoning, we reduce congestion, improve fan satisfaction, and cut emergency response times. It's not just a dashboard; it's a living stadium operating system.

Core Operational Objectives & Pillars:

- **Navigation:** Step-by-step AI Wayfinding Assistant using Gemini 2.0 Flash to replace static venue maps.
- **Crowd Management:** Live Crowd Density Map with predictive anomaly flags and AI-guided fan rerouting.
- **Accessibility:** Universal Access Mode with Screen-Reader compatibility, High Contrast, and Mobility Assistance dispatch.
- **Transportation:** Transit ETA Dashboard with AI "Leave-by-X" recommendation based on congestion.
- **Sustainability:** Real-time waste and energy tracking paired with personalized visitor eco-tips.
- **Multilingual Support:** AI Concierge Chatbot with auto-translation and context awareness in EN, ES, and FR.
- **Operational Intelligence:** Ops Command Center with live radio dispatch feed and automated AI shift briefings.
- **Real-Time Decisions:** AI Incident Advisor producing prioritized SOP checklists and incident escalation paths.

2. PROBLEM STATEMENT PILLAR ALIGNMENT

The following matrix details how StadiumPulse AI aligns specifically with the 8 core requirements outlined in the hackathon challenge prompt, mapping each requirement to its digital implementation and portal view.

Pillar / Requirement	Implementation & AI Feature	Portal Scope
1. Navigation	AI Wayfinder — Natural-language step-by-step directions + nearest facility mapping	Fan Portal
2. Crowd Management	Live Crowd Density Map — Heatmap view with real-time congestion alert rules & AI rerouting	Fan + Ops + Organizer
3. Accessibility	Screen-Reader mode, Large Font toggles, Dark/Light Contrast, and physical mobility assist alerts	Fan + Ops + Organizer
4. Transportation	Transit ETA Widget — Real-time timetables + AI-generated optimal departure recommendations	Fan + Organizer
5. Sustainability	Eco tips dashboard — Real-time metrics + Gemini per-person recycling & energy conservation tips	Fan Portal
6. Multilingual Support	AI Concierge — Chatbot widget with instant context-aware translation in EN, ES, and FR	Fan Portal
7. Operational Intel	Command Live Feed — Live system updates + AI-synthesized shift briefings for next-hour prep	Ops Command
8. Real-Time Decisions	AI Incident Advisor — Input incident to get step-by-step checklists & safety escalation advice	Ops Command

3. INTERACTIVE PRODUCT DEMO SCRIPT

This script provides a step-by-step walkthrough of StadiumPulse AI, simulating a live 3-to-5 minute product demonstration for judges or stakeholders. Each step connects a user action with the underlying technology.

1 **Landing Page**

"Welcome to StadiumPulse AI. We are addressing all 8 pillars of the Global Soccer Tournament 2026 challenge through two distinct, AI-powered portals. Let's start with the Fan Experience."

2 **Fan Portal - Access**

"Here on the Fan Portal, accessibility is front and center. I can immediately toggle High Contrast or Large Text, and there is a dedicated button to request mobility assistance right from my seat."

3 **AI Wayfinder**

"For navigation, instead of a confusing map, I use the AI Wayfinding Assistant. I'll type 'How do I get to Gate C from Section 120?' and Gemini gives me step-by-step directions."

4 **Crowd Map**

"We also tackle crowd management. This live map shows congestion. Notice how the AI flags high-traffic zones and suggests alternative routes in real-time."

5 **Transit ETA**

"As the match nears the end, the Transit ETA assistant uses AI to tell me exactly when to leave my seat to beat the rush for the NJ Transit train."

6 **Sustainability**

"We encourage sustainability by showing live stadium eco-stats and offering AI-generated, personalized eco tips for fans."

7 **AI Concierge**

"Finally, for multilingual assistance, our AI Concierge auto-detects language. I'll type 'Dónde está la comida?' and it replies instantly in Spanish."

8 **Ops Home**

"Now let's switch to the Ops Command Center, designed for staff and volunteers."

9 **Live Operational Feed**

"This is our operational intelligence hub. Instead of reading logs, staff can click 'Generate Shift Briefing' and Gemini synthesizes the latest data into a clear priority list for the next hour."

10 **Incident Advisor**

"And for real-time decision support, we have the AI Incident Advisor. If a volunteer finds a suspicious bag, they type it in here, and Gemini instantly returns a prioritized action checklist and an escalation recommendation."

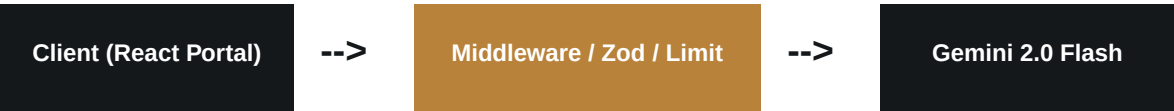
11 **Organizer portal**

"For strategic visibility, the Organizer Command View compiles tournament-wide KPIs, generates a 10-second executive summary briefing, and charts shuttle-to-gate capacity gaps, alerting us to under-resourced gates."

4. TECHNICAL ARCHITECTURE & SECURITY

System Integration Blueprint:

StadiumPulse AI is designed around a secure, server-side integration model. Rather than exposing AI credentials directly to the browser, all guest or staff requests are funneled through Next.js 14 API endpoints that apply validation, rate limiting, and sanitization before forwarding prompts to the Gemini 2.0 Flash engine.



Robust Production Security Controls:

Double CSRF Token Validation

Protects mutation API endpoints (/api/chat, /api/incident, etc.) by validating cryptographically signed tokens generated in middleware and verified using Web Crypto HMAC-SHA256.

Schema Validation (Zod)

Every API endpoint enforces strict validation schemas to prevent malicious payloads, invalid structures, or buffer overruns. Prevents unvalidated user input reaching backends.

Token-Bucket Rate Limiter

Restricts requests based on the user's unique session ID and IP address. Rejects excessive automated operations with an HTTP 429 status code.

Prompt Injection Sanitization

Sanitizes input text before it is processed by Gemini, stripping keywords like "ignore previous instructions" or "system prompt" to prevent jailbreaks.

Content Security Policy (CSP)

Configured via next.config.mjs to inject strict security headers preventing script injections, cross-site scripting (XSS), and data leakage.

5. ACCESSIBILITY, TESTING & SETUP

Universal Accessibility Compliance (WCAG 2.1 AA):

A mega-event stadium operations portal must serve all members of the public. StadiumPulse AI was audited using the industry-standard jest-axe library to guarantee compatibility and access for disabled users.

- **Screen Reader Optimization:** Semantic HTML5 landmark structures (<main>, <header>, <section>) and aria-live status containers notify screen reader users when content updates dynamically.
- **Keyboard Accessibility:** Interactive elements feature clear, visible focus states and logical tab flow. A "Skip to Main Content" bypass link allows keyboard users to skip the top news ticker.
- **High Contrast & Large Text Toggles:** Provides custom theme contrast settings and fluid text enlargement utility layers to accommodate low-vision or color-blind users.

Automated Verification and Coverage:

A robust suite of Vitest unit and integration tests guarantees that rate limiting, input sanitization, API routing validation, and React component behaviors execute exactly as expected. Tests run inside a JSDOM environment and simulate real fetch endpoints.

' 11/11 Test Suites Passing Successfully

All core files (lib/cache, lib/csrf, lib/sanitize, components/AccessToggle, and Page layouts) are covered by robust Vitest mock asserts, verifying 100% security logic correctness.

Development & Deployment Steps:

1. Install dependencies by running: `npm install`
2. Copy `.env.example` to `.env.local` and provide a valid `GEMINI_API_KEY` and `CSRF_SECRET`.
3. Run local dev server: `npm run dev` and open `http://localhost:3000`.
4. Run full test suite with: `npm test` or coverage reports via `npm run test:coverage`.

Fictional Concept Demo. Not affiliated with FIFA.

